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SARDAR PATEL UNIVERSITY

S.Y.B.C.A. (Semester-III) EXAMINATION November- 2017

US03EBCA02 : Introduction to Artificial Intelligence

Date: 17/11/2017

Time: 02:00 to 04:00

Total Marks : 70

Q.1 Multiple choice questions:

[10]

1. Medical diagnosis is a _____.
A. Mundane task B. Formal task C. Expert task D. None of these
2. _____ is considered to be the first AI program.
A. The Logic Theorist B. DENDRAL
C. MYCIN D. ELIZA
3. Inferring facts from existing data is also known as _____.
A. Reasoning B. Retrieving
C. Learning D. Acquisition
4. In _____, each rule contains two confidence estimates.
A. MYCIN B. DENDRAL C. R1 D. PROSPECTOR
5. _____ use symbolic representations for knowledge.
A. Expert system B. Expert shell
C. Inference engine D. Knowledge base
6. Which of the following will find the best goal state?
A. Best-first search B. Depth-first search
C. Bidirectional search D. Breadth-first search
7. Predicates are also known as _____.
A. Functions B. First-order
C. Atoms D. Heuristic
8. The terms Analog signals, Real time, Sensors and special hardware are used for _____.
A. Computerised gaming B. Robots
C. Language translators D. None of these
9. A _____ can be regarded as a label applied to a linguistic concept which has no precise boundary.
A. Fuzzy Logic B. Fuzzy set C. Crisp set D. None of these
10. The conventional machine uses _____.
A. Fuzzy Logic B. Fuzzy set C. Crisp set D. None of these

(P.T.O.)

Q.2 Write short answer of any ten.

[20]

1. What is ELIZA?
2. List the events that took place in 1956
3. List the Task domains of AI
4. State the stages of knowledge acquisition
5. What is a Knowledge base?
6. What is a User interface?
7. What is interactive deepening depth first search
8. State the connectives of FOPL
9. What is Depth-first search
10. State any two problems with Robots.
11. What is Fuzzy logic
12. State any 2 reasons why game playing was appealing for AI researchers

Q.3(a) Explain history of AI in brief.

[5]

(b) Write a note on AI Problem.

[5]

OR

Q.3(a) What is AI? Explain in Brief.

[5]

(b) Write a note on the underlying assumption of AI problem.

[5]

Q.4(a) Write a note on Knowledge acquisition.

[5]

(b) Write a note on the characteristic features of an expert system.

[5]

OR

Q.4(a) Write a note on knowledge representation.

[5]

(b) What is an Expert shell? Write a note on Expert shell.

[5]

Q.5 Write a note on predicate logic.

[10]

OR

Q.5 Explain following:

[10]

1. depth first search
2. breadth first search
3. depth limited search
4. interactive deepening depth first search
5. bidirectional search.

Q.6(a) Write a note on Fuzzy logic.

[6]

(b) Write a note on Game playing.

[4]

OR

Q.6(a) Write a note on crisp sets.

[6]

(b) Write a note on Robotics.

[4]

—X—